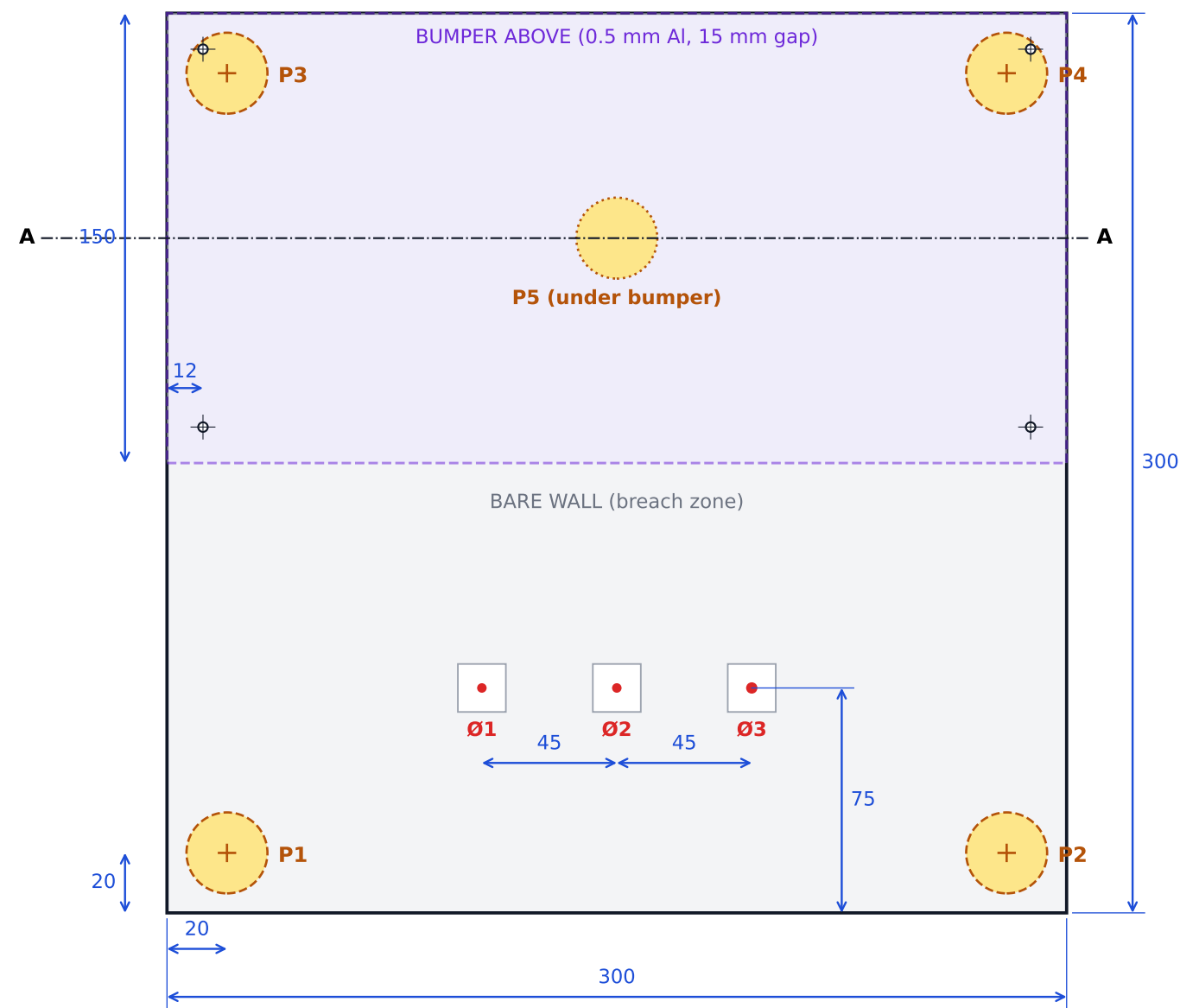
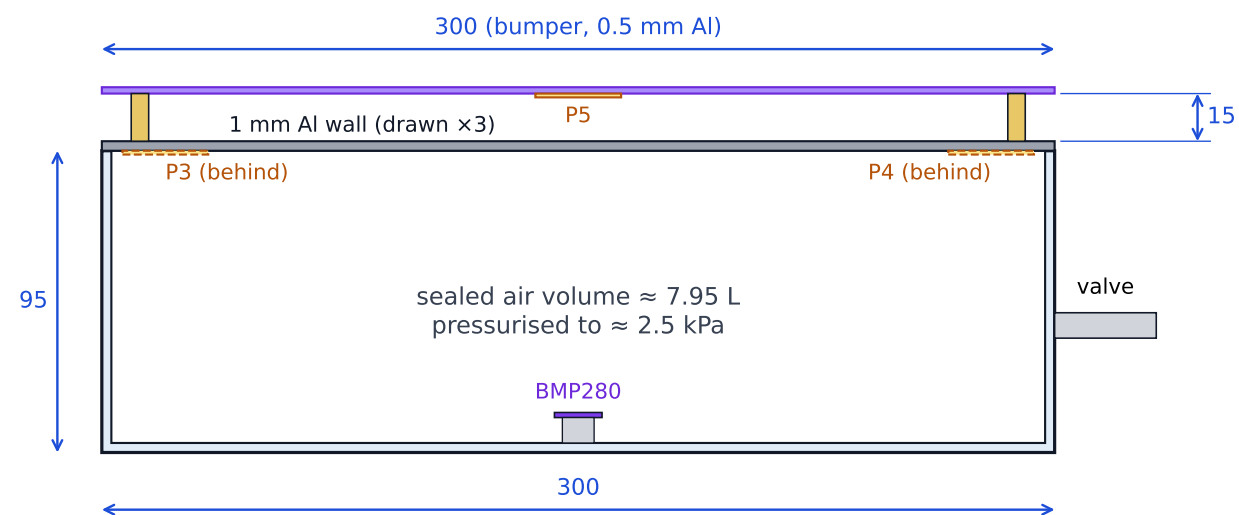


TOP VIEW · WALL PLATE LAYOUT (not to scale; dimensions govern)



Breach ports are drilled, then sealed with foil and tape; a punch through the foil opens a hole of known diameter. Piezos P1–P4 sit 20 mm in from each corner on the underside of the wall; P5 sits on the underside of the bumper.

SECTION A-A (cut at $y = 225$ through P5, viewed from $-y$)



#	Part	Spec	Qty
1	Pressure wall	Aluminium sheet, 1 mm, 300 × 300	1
2	Bumper (Whipple shield)	Aluminium sheet, 0.5 mm, 300 × 150	1
3	Sealed box	Plastic container ~300 × 300 × 95, ~7.9 L	1
4	Piezo disc	27 mm brass, bonded with Fevikwik	5
5	Standoff + screw + nut	M3 × 15 hex, brass	4
6	Pressure sensor	BMP280 module, I ² C 0x76	1
7	Valve	Tubeless tyre valve stem	1
8	Seal	M-seal epoxy bead + foil over ports	—
9	Analog board	Perfboard: 2× LM324, 2× LM339, 5 trimpots	1
10	Controller	ESP32 30-pin dev board	1

HULLSENSE · Strike-localisation and breach-sizing test rig

Team BYTEMELATER · Team lead: Yogeeshwaran C · Rajalakshmi Engineering College · SEDHACKS '26

All dimensions in mm · Sheet A3 · Source: hardware/cad/hullsense_rig.py (CadQuery) · STEP: hullsense_rig.step

Sensor coordinates match docs/dashboard/solver.js. Wall thickness exaggerated in section for clarity.